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// OSSP Skill 23 - Large Pipelines and Multiple Jobs
// Demonstrates a longer pipeline, multiple child processes,
// stability/error monitoring, synchronization, and timing.

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>

#define STAGES 4

static void die(const char *msg) {
    perror(msg);
    exit(EXIT_FAILURE);
}

int main(void) {
    printf("=== Skill 23: Large Pipeline + Multiple Processes ===\n");
    printf("Pipeline: printf -> tr -> sort -> wc\n");

    int pipes[STAGES - 1][2];

    for (int i = 0; i < STAGES - 1; i++)
        if (pipe(pipes[i]) == -1)
            die("pipe");

    pid_t pids[STAGES];
    clock_t start = clock();

    for (int i = 0; i < STAGES; i++) {
        pids[i] = fork();

        if (pids[i] == -1)
            die("fork");

        if (pids[i] == 0) {
            if (i > 0)
                dup2(pipes[i - 1][0], STDIN_FILENO);
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    if (i < STAGES - 1)
        dup2(pipes[i][1], STDOUT_FILENO);

    for (int j = 0; j < STAGES - 1; j++) {
        close(pipes[j][0]);
        close(pipes[j][1]);
    }

    if (i == 0)
        execlp("printf", "printf",
               "banana\napple\norange\napple\n", NULL);
    else if (i == 1)
        execlp("tr", "tr", "a-z", "A-Z", NULL);
    else if (i == 2)
        execlp("sort", "sort", NULL);
    else
        execlp("wc", "wc", "-l", NULL);

    perror("exec");
    _exit(127);
}

for (int i = 0; i < STAGES - 1; i++) {
    close(pipes[i][0]);
    close(pipes[i][1]);
}

int failures = 0;

for (int i = 0; i < STAGES; i++) {
    int status;

    if (waitpid(pids[i], &status, 0) == -1) {
        perror("waitpid");
        failures++;
        continue;
    }
}
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    if (!WIFEXITED(status) || WEXITSTATUS(status) != 0)
        failures++;
}

double elapsed = (double)(clock() - start) / CLOCKS_PER_SEC;

printf("\nAll %d processes synchronized.\n", STAGES);
printf("Failures detected: %d\n", failures);
printf("Measured CPU time: %.6f seconds\n", elapsed);

return failures ? 1 : 0;
}
```

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sai_pavani@Sai-Pavani:~/OS_SKILL$ nano skill_23.c.c
sai_pavani@Sai-Pavani:~/OS_SKILL$ gcc skill_23.c.c -o skill_23
sai_pavani@Sai-Pavani:~/OS_SKILL$ ./skill_23
=== Skill 23: Large Pipeline + Multiple Processes ===
Pipeline: printf -> tr -> sort -> wc
4

All 4 processes synchronized.
Failures detected: 0
Measured CPU time: 0.001039 seconds
sai_pavani@Sai-Pavani:~/OS_SKILL$ |
```